DNPH Assay

Overview

Bacterial culture supernatant is rapidly screened for ACC deaminase (ACCD) enzymatic activity using the 2,4-dinitrophenylhydrazine (DNPH) colorimetric assay. The assay shows whether functional ACCD is actively cleaving the ACC substrate into α-ketobutyrate, determining net enzyme activity, standard curve calibration, and total product accumulation across engineered strains.

Materials and Methods

- Regents and solutions required for the experiment:
    1. 2,4-Dinitrophenylhydrazine(DNPH) Solution (0.2% w/v in 2N HCl):
       Dissolve 0.20g of 2,4-Dinitrophenylhydrazine powder in 100 mL of 2 N HCl solution. Protect solution from light exposure by covering the container with aluminum foil and store and 4 °C. Handle 2 N HCl with proper personal equipment (PPE), including acid-resistance gloves and safety goggles, and always operate under a chemical fume cover due to corrosive hazard.

    2. Hydrochloric Acid (0.56 H NaOH):
       Prepare by diluting concentrated HCl (12N) with ultrapure water used specifically for enzymatic quenching and stabilization for acid-catalyzed reaction .

    3. Sodium Hydroxide Solution (2N NaOH):
       Dissolve 8.0 g of NaOH pellets in deionize water to bring the total volume to 100 mL. This step will cause an exothermic reaction, so always operate this step in the ice bath.

    4. 0.5 M ACC substrate solution:
       Dissolve 50.55 mg ACC per 1.0 mL deionized water and store at -20 °C and pre-equilibrate at 30 °C for at least 3 min before use.

    5. 20X reaction buffer:
       1 M phosphate buffer pH 8.5 containing 250 μM PLP buffer (store at 4 °C without light exposure, the solution must be used within 2 weeks)

    6. α-Ketobutyrate primary standard stock (10 mM):
       Dissolve 12.4 mg Sodium 2-oxobutryrate (MW 124.7) per 10 mL water (prepare on ice before using daily)

- Experiment protocol:

1. Mix 100 µL filtered cultural supernatant with 12 µL 20X reaction buffer and 10 µL of 0.5 M ACC substrate solution, incubate at 30 °C and react at least overnight.

2. Transfer 10 µL of reaction mixture into 50 µL of 0.56 N HCl in a 1.5 mL microtube and vortex immediately to quench the precipitant protein.

3. Centrifuge at 12,000 xg for 5 min at RT, then transfer 50 µL of clear supernatant into a new clean microtube.
4. Dilute the α-ketobutyrate by the following table:

Diluted concentration (mM)	Final concentration (mM)	ddH_2O (µL)	α-ketobutyrate
50	4.098	100	+100 µL, 100 mM
25	2.049	100	+100 µL, 50 mM
12.5	1.025	100	+100 µL, 25 mM
6.25	0.512	100	+100 µL, 12.5 mM
3.125	0.256	100	+100 µL, 6.125 mM
0	0	200	0

5. Prepare α-ketobutyrate standard. 50 µL supernatant add 6 µL 1 M phosphate buffer pH 8.5 containing 250 µM PLP buffer and 5 µL diluted α-ketobutyrate

6. Add 40 µL of 0.56 N HCl and 15 µL of 0.2% DNPH/2 N HCl solution, then vortex and incubate the solution at 30 °C for 30 min.

7. Add 100 µL of 2 N NaOH solution, vortex it for 10 sec, and transfer the mixed solution (with a total volume of 205 µL) into a 96-well flat bottom plate. Then measure the A540 after 10 min.

Calculate

The calculation can be break down into few steps as shown below:

1. Net absorbance Evaluation

In order to correct the number for background interference, simply subtract the absorbance of the HCl (sample blank) from the DNPH-treated sample at the peak absorbance wavelength, shown as below:

$\Delta A = A_{sample} - A_{blank}$

2. Standard Curve Interpolation
   The concentration of α-ketobutyrate in the sample can be determined by using the linear regression equation :

$$y = m \cdot x + b$$

   The equation above than can be rearranged to find the concentration:

$$\text{Concentration of alpha-ketobutyrat(mM)} = \Delta A - b/m$$

   Where:
   ΔA: Net Absorbance of the sample (y)
   m: Slope of the standard curve
   B: y-intercept of the standard curve

3. Quantitation of Total Production:
   Apply the equation below to convert the concentration into total accumulated product (nmol) within the total reaction assay volume:

   Total Product (nmol)=Concentration of alpha-ketobutyrate(mM) * Vreaction (μL)

   Where:
   Vreaction: Total Volume of the assay reaction mixture